II. Physikalisches Anfängerpraktikum

UNIVERSELLE HEIDELBEERE
EIGENTLICH EINE BROOMBEERE

Protocol for the Experiment

X-Ray

(Experiment 255)

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FEZ

Preface and Notes

Use of External Tools

For the linguistic drafting and correction of the Introduction (Kapitel **??**) and the Execution (Kapitel 3.), the AI *Lumo AI* as well as *Duden Mentor* were employed. However, the substantive evaluation and the actual creation of the document were carried out entirely without the assistance of Artificial Intelligence, unless explicitly marked otherwise. [Dud26, Pro26]

All graphics contained in the document that lack a specific reference were independently created or edited in Inkscape, or are plots generated in Python. Both the Python code and the LaTeX document were prepared in the Visual Studio Code (VS Code) development environment.

Own Code and Transparency

To carry out the evaluations, a custom Python library was developed. The entire source code - both the Python scripts and the LaTeX sources - is self-created and publicly available online.

The entire project structure can be found on my GitHub. Here, both the Python code and the LaTeX document code are documented. [Fin26a]

The Python functions used are specifically identifiable on GitHub under the name „Papulator“. [Fin26b]

Use of University Materials

This document is based on the script by Jens Wagner (as of April 27, 2026). Graphics from this script were adopted for the Introduction. These excerpts are transparently marked; the figures used always originate from the chapters of the script that share the same experiment number and experiment name as in the present document. In this experiment, the graphics from *Experiment 255 X-Ray* were used. [Wag26]

Table of Content

1. Introduction

1.1 Task and Motivation

In this experiment, the generation and analysis of X-ray radiation are investigated. The aim is, on the one hand, to determine fundamental constants of physics experimentally and, on the other hand, to analyze the structure of crystals using diffraction methods. Specifically, Planck's quantum of action h is to be determined from the short-wavelength end of the bremsstrahlung spectrum as well as from the threshold voltage at characteristic wavelength. Furthermore, the wavelengths of the characteristic K_α and K_β lines of a molybdenum anode are to be determined. By using lithium fluoride (LiF) and sodium chloride (NaCl) as diffraction crystals, the connection between crystal lattice structure and macroscopic quantities such as the Avogadro number N_A can also be established.

1.2 Physical Basis

1.2.1 Generation of X-ray Radiation

The generation of X-ray radiation in this experiment takes place via an X-ray tube with a molybdenum anode and a heated filament cathode. Inside an evacuated glass vessel, electrons are accelerated by an electric field between the cathode and the anode. The kinetic energy E of the electrons results from the applied accelerating voltage U:

$$E = eU \tag{1.1}$$

where e represents the elementary charge. Upon impact with the anode, the electrons are decelerated in the Coulomb field of the atomic nuclei. This braking process leads to the emission of a continuous spectrum of electromagnetic radiation, known as the bremsstrahlung spectrum. The maximum energy of an emitted photon is reached when an electron converts its entire kinetic energy into a single photon. This results in a minimum possible wavelength λ_{gr}, the cutoff wavelength (or short-wavelength limit), which is described by conservation of energy:

$$eU = h\nu_{gr} = \frac{hc}{\lambda_{gr}} \tag{1.2}$$

Here, h is Planck's quantum of action and c is the speed of light. In addition to the continuous spectrum, a characteristic line spectrum is generated if the voltage is sufficiently high. This occurs when electrons are knocked out of inner shells of the anode material and the resulting vacancies are filled by electrons from higher shells. For the transition from the n-th to the m-th shell, the energy $E_{n\to m}$ can be estimated according to Moseley's Law:

$$E_{n\to m} = hcR_\infty(Z-A)^2\left(\frac{1}{m^2}-\frac{1}{n^2}\right) \tag{1.3}$$

Here, R_∞ denotes the Rydberg constant, Z the atomic number of the anode material (here molybdenum), and A a shielding constant, which is approximately $A \approx 1$ for K_α radiation ($n = 2 \to m = 1$).

1.2.2 Bragg Reflection

To measure the wavelengths of the generated X-ray radiation, crystals are used as diffraction gratings. Since the wavelengths of X-ray radiation are of the same order of magnitude as the interatomic distances in crystals, constructive interference occurs at the lattice planes of the crystal. This phenomenon is called Bragg reflection. An incident ray is reflected at various parallel lattice planes. For these reflected rays to overlap constructively, the path difference Δs must be an integer multiple of the wavelength λ. Bragg's Law describes this relationship:

$$2d\sin(\vartheta) = n\lambda, \quad n \in \mathbb{N} \tag{1.4}$$

Here, d is the distance between adjacent lattice planes, ϑ is the angle of incidence (glancing angle), and n is the order of reflection. By varying the angle ϑ and measuring the intensity, the X-ray spectrum can be recorded.

1.2.3 Crystals and Lattice Structure

The LiF and NaCl crystals used in the experiment both possess a face-centered cubic structure. In this structure, a unit cell repeats periodically in space. There are four atoms (or ion pairs) per unit cell. For a cut along the principal planes, the lattice plane spacing d in terms of the lattice constant a is given by:

$$d = \frac{a}{2} \tag{1.5}$$

If the wavelength λ is known, the lattice constant a can be determined from the measured angle ϑ. Conversely, a known lattice constant enables the determination of microscopic quantities from macroscopically measurable properties. The Avogadro number N_A, which indicates the number of particles per mole, can be calculated via the volume of the unit cell. The volume of a unit cell is $V = (2d)^3$. With the molar mass M_{mol} and the density ρ of the crystal, it follows that:

$$N_A = \frac{4V_{mol}}{(2d)^3} = \frac{4M_{mol}}{\rho(2d)^3} = \frac{1}{2}\frac{M_{mol}}{\rho d^3} \tag{1.6}$$

This relationship allows the Avogadro number to be determined indirectly via X-ray diffraction. In the evaluation, the statistical nature of radioactive or count-rate-like events is also taken into account. Since the count rate M follows a Poisson process, the standard error of a single measurement is:

$$\sigma = \sqrt{M} \tag{1.7}$$

This value is used to incorporate the uncertainty of the measured intensities into the error propagation.

3. Experimental Procedure

3.1 Measurement Method

The experiment is based on the rotating crystal method for spectrometry. In this approach, a crystal holder equipped with the respective detection crystal (initially LiF, later NaCl) is positioned in the X-ray beam path. A Geiger-Müller counter tube serves as the detector for intensity-measured X-ray radiation. The actual measurement process is performed automatically by a control program on the computer. First, a safety check of the device is carried out, ensuring that the lead-glass window is closed before the high voltage is activated.

For recording the bremsstrahlung spectrum, the crystal is rotated over a specific angular range while, for each angular position, the count rate of detected photons is measured over a predetermined time period. To increase accuracy in determining the characteristic lines (K_α and K_β), after a coarse preliminary scan, detailed measurements are performed only in the regions of expected maxima. Here, the step size is reduced to capture the shape of the peaks precisely.

Another part of the experiment investigates the relationship between the count rate and the accelerating voltage of the X-ray tube at a constant detection angle. Here, the voltage is varied stepwise while the count rate is registered for each setting. Finally, the entire measurement sequence is repeated with an NaCl crystal to determine the lattice constant using the already determined wavelengths, and from this, the Avogadro number. All measured values are recorded digitally and documented directly.

3.2 Experimental Setup

The experimental setup consists of three main components: the X-ray source, the goniometric detection system, and the evaluation electronics. The X-ray tube features a water-cooled copper holder and a molybdenum anode, arranged inside the vacuum glass bulb opposite a heated tungsten cathode. A high-voltage source generates the electric field for accelerating the electrons.

The detection system includes a precision goniometer, on which the crystal holder and the counter tube are mounted movably. The rotation of the crystal and the simultaneous movement of the counter occur synchronously according to the Bragg angle (1:2 relation between crystal and counter rotation), so that the detector always captures the reflected beam. The counter tube is connected to an amplifier and a pulse counter, which digitizes the signals and forwards them to a computer. For safety, the entire beam path is embedded in a lead housing, accessible through a swivel lead-glass window. An internal safety switch ensures that the high voltage is switched off immediately as soon as the window is opened.

Sketch of the Experimental Setup

Abbildung 3.I:
Schematic setup of the X-ray spectrometer following the rotating crystal method.

4. Evaluation

Uncertanty Calculation

For the statistical evaluation of n messurments x_i, the follwoing equations are defined: [Wag25]:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \text{Arithmetic Mean} \tag{4.1}$$

$$\sigma^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad \text{Variance} \tag{4.2}$$

$$\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} \qquad \text{Standard Deviation} \tag{4.3}$$

$$\Delta\bar{x} = \frac{\sigma}{\sqrt{n}} = \sqrt{\frac{1}{n(n-1)}\sum_{i=1}^{n}(\bar{x} - x_i)^2} \qquad \text{Uncertainty of the Mean} \tag{4.4}$$

$$\Delta f = \sqrt{\left(\frac{\partial f}{\partial x}\Delta x\right)^2 + \left(\frac{\partial f}{\partial y}\Delta y\right)^2} \qquad \text{Gaussian Error Propagation for } f(x,y) \tag{4.5}$$

$$\Delta f = \sqrt{(\Delta x)^2 + (\Delta y)^2} \qquad \text{Uncertainty for } f = x + y \tag{4.6}$$

$$\Delta f = |a|\Delta x \qquad \text{Uncertainty for } f = ax \tag{4.7}$$

$$\frac{\Delta f}{|f|} = \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2} \qquad \text{Relative Uncertainty for } f = xy \text{ or } f = x/y \tag{4.8}$$

$$\sigma = \frac{|a_{lit} - a_{gem}|}{\sqrt{\Delta a_{lit}^2 + \Delta a_{gem}^2}} \qquad \text{Significant Deviation Calculation} \tag{4.9}$$

4.1 X-Ray Spectrum with Lithium-Florid

In the first part, the X-Ray spectrum with the mounted Lithium-Florid (LiF) cirstal is being evaluated. This spectrum is used to estimate Planck's constant h, evaluate the wavelengths for the peaks – the characteristic lines – and find the influence the acceleration voltage has on the estimated Planck's constant.

4.1.1 Calculating the Plank's Constant

In this section, the paper will use the messurment data, and set a regression to the (nearly) linear part on the beginning. For the angle an uncertainty of $\Delta\vartheta = 0.05°$ and for the Counts we use equation 4.3. The linear regression $f(\vartheta) = m \cdot \vartheta + b$ has a slope m and corsses the ordinate-axis at b. Both values can be found in the plot 4.I.

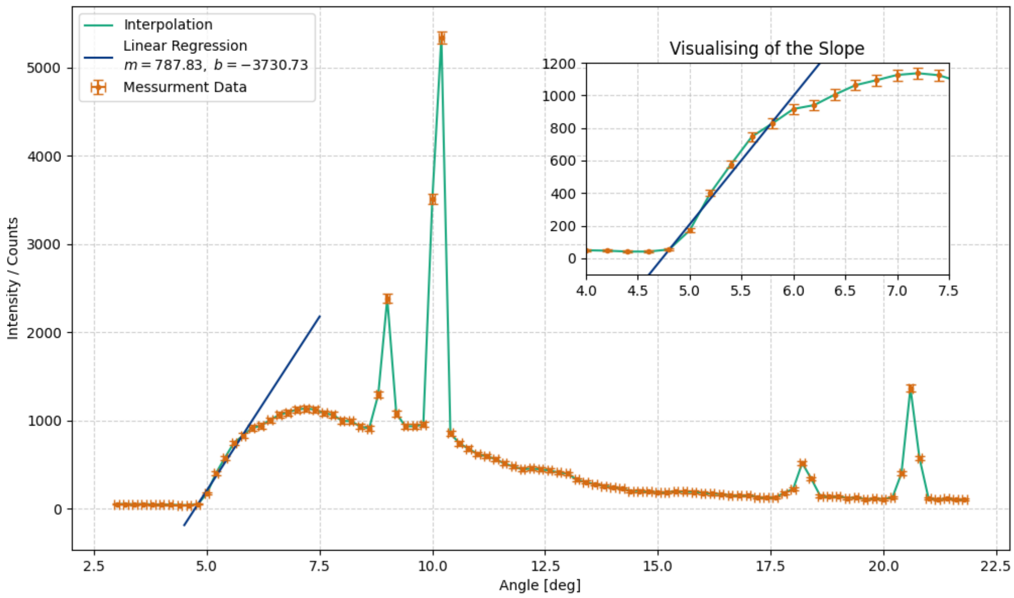

Abbildung 4.I:
Messurment data interpolated along side the linear regression.

By setting the linear regression equal to zero and solve for ϑ, we reseve the critical angle:

$$\vartheta_{\text{crit}} = -\frac{b}{a} \tag{4.10}$$

Using equation 1.4, using $n = 1$, $d = 201.4\text{pm}$ and the just calculated angle ϑ_{crit} the wavelength is calculated to be

$$\lambda = \tag{4.11}$$

4.1.2 Evaluation of the K-Lines

4.1.3 Counts in dependence of the acceleration voltage

4.2 X-Ray Spectrum with Sodium-Chlorid

4.2.1 Lattice of the NaCl Cristal

4.2.2 Evaluation of the Avogadro-Number

5. Diskussion

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Abbildungsverzeichnis

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Literaturverzeichnis

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